

Inner Product Space

→ consider a vector space $\mathbb{R}^n$. Prove that $\mathbb{R}^n$ is an inner product space which means

$$\langle u, v \rangle = \sum_{i=1}^n u_i v_i \rightarrow \text{standard inner product}$$

Is it binary operation?

$$\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \quad (\because \text{not } \mathbb{R}^n, \text{ therefore not binary operation})$$

Inner product can be defined in any way it just needs to satisfy the following conditions

① linearity $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
 $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$

②

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$$

Proof: let $\underline{u}, \underline{v}, \underline{w}$ be vectors in vector space $\mathbb{R}^n$

$$\downarrow$$
$$\left\{ \begin{array}{l} \underline{u} = [u_1, u_2, u_3, \dots, u_n]^T \\ \underline{v} = [v_1, v_2, \dots, v_n]^T \\ \underline{w} = [w_1, w_2, \dots, w_n]^T \end{array} \right.$$

$$\text{Let } \underline{z} = \underline{u} + \underline{v} = [u_1 + v_1, u_2 + v_2, \dots, u_n + v_n]^T$$

$$\begin{aligned} \langle \underline{u} + \underline{v}, \underline{w} \rangle &= \langle \underline{z}, \underline{w} \rangle = \sum z_i w_i \\ &= \sum (u_i + v_i) w_i \\ &= \sum u_i w_i + \sum v_i w_i \\ &= \langle \underline{u}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle \end{aligned}$$

$$\text{Let } \underline{u} = \alpha \underline{u}$$

$$\underline{u} = [\alpha u_1, \alpha u_2, \dots, \alpha u_n]^T$$

$$\langle \alpha \underline{u}, \underline{v} \rangle = \sum \alpha u_i v_i = \alpha \sum u_i v_i = \alpha \langle \underline{u}, \underline{v} \rangle$$

(ii) Symmetry $\langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle$

(iii) Positive Definiteness $\langle \underline{u}, \underline{u} \rangle \geq 0$

$$\langle \underline{u}, \underline{u} \rangle = \sum u_i^2 \geq 0 \quad [\text{Real number square is always greater than } 0]$$

$$= 0 \quad \text{if \& only if } \underline{u} = 0$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} y+z \\ x+2y+z \\ x+y+z \end{bmatrix}$$

$$t \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Consider a vector space $\mathbb{R}^n$. Prove that $\mathbb{R}^n$ is inner product space with inner product defined as

$$\mathbb{R}^n, \langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^n u_i v_i \quad (\text{standard inner product for real inner system}).$$

Is it binary operation?

$$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R} \quad (\because \text{not } \mathbb{R}^n, \text{ therefore not binary operation})$$

Inner product can be defined in any way it just needs to satisfy the following conditions:-

$$\textcircled{1} \text{ Linearity} \rightarrow \langle \underline{u} + \underline{v}, \underline{w} \rangle = \langle \underline{u}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle$$

$$\langle \alpha \underline{u}, \underline{v} \rangle = \alpha \langle \underline{u}, \underline{v} \rangle$$

or

$$\langle \alpha \underline{u} + \beta \underline{v}, \underline{w} \rangle = \alpha \langle \underline{u}, \underline{w} \rangle + \beta \langle \underline{v}, \underline{w} \rangle$$

Proof:

let $\underline{u}, \underline{v}, \underline{w}$ be vectors in vector space $\mathbb{R}^n$.

$$\rightarrow \left. \begin{aligned} \underline{u} &= [u_1, u_2, \dots, u_n]^T \\ \underline{v} &= [v_1, v_2, \dots, v_n]^T \\ \underline{w} &= [w_1, w_2, \dots, w_n]^T \end{aligned} \right\}$$

$$\text{let } \underline{z} = \underline{u} + \underline{v} = [u_1 + v_1, u_2 + v_2, \dots, u_n + v_n]^T$$

$$\langle \underline{u} + \underline{v}, \underline{w} \rangle = \langle \underline{z}, \underline{w} \rangle = \sum z_i w_i$$

$$= \sum (u_i + v_i) w_i$$

$$= \sum u_i w_i + \sum v_i w_i$$

$$= \langle \underline{u}, \underline{w} \rangle + \langle \underline{v}, \underline{w} \rangle$$

Let, $\underline{u} = \alpha \underline{u}$

$$\underline{u} = [\alpha u_1, \alpha u_2, \dots, \alpha u_n]^T$$

$$\langle \alpha \underline{u}, \underline{v} \rangle = \sum \alpha u_i v_i = \alpha \langle \underline{u}, \underline{v} \rangle$$

(ii) Symmetry : $\langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle$

(iii) Positive Definiteness : $\langle \underline{u}, \underline{u} \rangle \geq 0$

$$\langle \underline{u}, \underline{u} \rangle = \sum_{i=1}^n u_i^2 \geq 0 \quad (u_i^2 \geq 0; \text{ real number square is always greater than 0})$$

$= 0$ if & only if $\underline{u} = 0$!

Ex:-

$$\underline{u} = \begin{bmatrix} 2+2i \\ 5 \end{bmatrix}, \quad \underline{v} = \begin{bmatrix} 3i \\ 5 \end{bmatrix}$$

$$\langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^2 u_i v_i = (2+2i)3i + 25 = 6i - 6 + 25 = 6i + 19$$

$$\langle \underline{v}, \underline{u} \rangle = \langle \underline{u}, \underline{v} \rangle = 19 + 6i$$

Ex:-

$$\underline{u} = \begin{bmatrix} 2+2i \\ 3 \end{bmatrix}, \quad \underline{v} = \begin{bmatrix} 3i \\ 5 \end{bmatrix}, \quad \underline{w} = \begin{bmatrix} 3 \\ 5i \end{bmatrix}$$

$$\langle \underline{u}, \underline{u} \rangle = (2+2i)(2-2i) + 9$$

$$= 4 + 8i - 4 + 9$$

$$\langle \underline{v}, \underline{v} \rangle = 25 - 9 = 16$$

$$\langle \underline{w}, \underline{w} \rangle = -25 + 9 = -16 \quad (\text{positive definiteness is not satisfied})$$

Since we cannot change the rules, so we need to change the formula

Standard Inner Product : $\langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^n u_i \overline{v_i}$

$$\langle \underline{u}, \underline{u} \rangle = (2+2i)(2-2i) + 9$$

$$= 4 + 4 + 9 = 17$$

$$\langle \underline{v}, \underline{v} \rangle = (3i)(-3i) + 25$$

$$= 9 + 25 = 34$$

$$\langle \underline{w}, \underline{w} \rangle = 9 + 25$$

$$= 34$$

In this case the 3rd condition is always satisfied. ✓

So finally, conditions are as follows

1. Linearity : $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$
2. Symmetry : $\langle u, v \rangle = \overline{\langle v, u \rangle} \rightarrow$ overbar represents complex conjugate
3. Positive Definiteness : $\langle u, u \rangle \geq 0$

$$\begin{aligned} \langle u, \alpha v \rangle &= \langle \alpha v, u \rangle = \overline{\alpha \langle v, u \rangle} = \overline{\alpha} \overline{\langle v, u \rangle} = \overline{\alpha} \langle u, v \rangle \end{aligned}$$

$$\langle u, v \rangle = \sum_{i=1}^m u_i \overline{v_i}$$

$$\langle f, g \rangle = \int_a^b f(x) \overline{g(x)} dx$$

Ex:- Determine the value of λ which makes the following operation inner product -
 $\underline{x}, \underline{y} \in \mathbb{R}^2$

$$\langle \underline{x}, \underline{y} \rangle = x_1 y_1 - 3x_1 y_2 - 3x_2 y_1 + \lambda x_2 y_2$$

for this to be inner product

$$\langle \underline{x}, \underline{x} \rangle \geq 0$$

$$\langle \underline{x}, \underline{x} \rangle = x_1^2 - 3x_1 x_2 - 3x_2 x_1 + \lambda x_2^2 \geq 0$$

$$x_1^2 - 6x_1 x_2 + \lambda x_2^2 \geq 0$$

$$\left(\frac{x_1}{x_2}\right)^2 - 6\left(\frac{x_1}{x_2}\right) + \lambda \geq 0 \Rightarrow y^2 - 6y + \lambda \geq 0$$

for this to be true $b^2 - 4ac \leq 0$

$$36 - 4\lambda \leq 0$$

$$4\lambda \geq 36$$

$$\lambda \geq 9$$

Ex:- Determine a non-zero vector in $\mathbb{R}^3$ which is orthogonal to $\underline{v}_1, \underline{v}_2, \underline{v}_3$.

$$\underline{v}_1 = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix}^T$$

$$\underline{v}_2 = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix}^T$$

$$\underline{v}_3 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$$

$$\text{let } \underline{x} = \begin{bmatrix} a & b & c \end{bmatrix}^T$$

$$\langle \underline{x}, \underline{v}_1 \rangle = 0$$

$$a + b + 2c = 0$$

$$\langle \underline{x}, \underline{v}_2 \rangle = 0$$

$$2a + b + 3c = 0$$

$$\langle \underline{x}, \underline{v}_3 \rangle = 0$$

$$a + 2b + 3c = 0$$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 2 & 1 & 3 & 0 \\ 1 & 2 & 3 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} a + b + 2c &= 0 \\ b + c &= 0 \end{aligned}$$

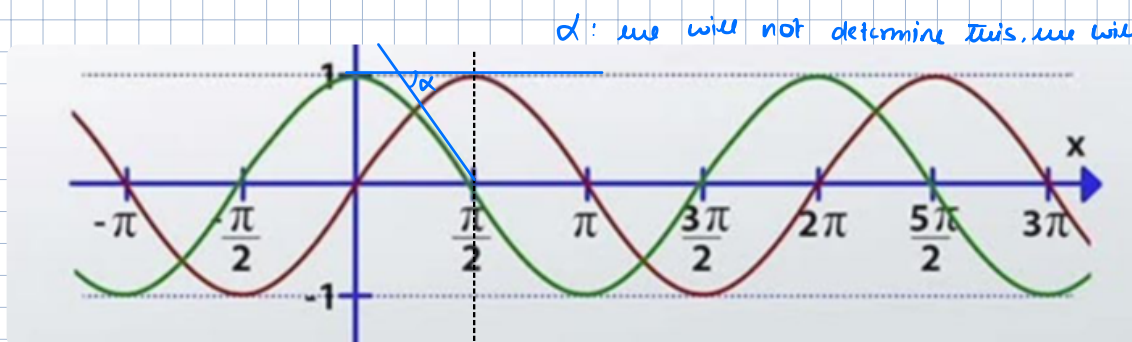
$$\text{DoF} = 3 - 2 = 1 \quad \Phi \quad c = \alpha, b = -\alpha, a = \alpha$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$\therefore$ we have a vector $[1, -1, 1]^T$ & every displacement is orthogonal.

(Q.) Verify the following functions are orthogonal

- $\sin x, \cos x$ in $[-\pi, \pi]$
- $\sin n\pi x, \sin m\pi x$ in $[-1, 1]$
- $\sin n\pi x, \cos m\pi x$ in $[-1, 1]$
- $\cos n\pi x, \cos m\pi x$ in $[-1, 1]$



$$\therefore \langle f(x), g(x) \rangle = \int_a^b f(x) \overline{g(x)} dx$$

a) $f(x) = \sin x, g(x) = \cos x$

$$\langle f(x), g(x) \rangle = \int_{-\pi}^{\pi} \sin x \cos x dx = \frac{1}{2} \int_{-\pi}^{\pi} \sin 2x dx = 0$$

b) $f(x) = \sin n\pi x, g(x) = \sin m\pi x$ in $[-1, 1]$

$$\langle f(x), g(x) \rangle = \int_{-1}^1 \sin(n\pi x) \sin(m\pi x) dx$$

$$= \frac{1}{2} \int_{-1}^1 [\cos(m-n)\pi x - \cos(m+n)\pi x] dx.$$

$$= \frac{1}{2} \left\{ \frac{\sin(m-n)\pi x}{(m-n)\pi} - \frac{\sin(m+n)\pi x}{(m+n)\pi} \right\}_{-1}^1$$

How this becomes 0?

Solve (b), (c), (d) part by yourself

(#) $f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x)$ Fourier Series Expansion

Taking inner product with $\sin(m\pi x)$ from -1 to $+1$

$$\langle f(x), \sin(m\pi x) \rangle = \int_{-1}^1 \sum_{n=1}^{\infty} a_n \sin(n\pi x) \sin(m\pi x) dx$$

(Q.) Let B be the vector space of all the polynomial over $\mathbb{R}$ degree less than or equal to 2.

Determine a polynomial which is orthogonal to $f(t)$ if

$$\langle f, g \rangle = \int_0^1 f(t) g(t) dt \quad f(t) = 2t+1$$

Sol'n: Let $g(t) = a_0 + a_1 t + a_2 t^2$

Now determining a_0, a_1 and a_2 such that -

$$\int_0^1 (a_0 + a_1 t + a_2 t^2) (2t+1) dt = 0$$

$$\int_0^1 (2a_0 t + a_0 + 2a_1 t^2 + a_1 t + 2a_2 t^3 + a_2 t^2) dt = 0$$

$$\int_0^1 (2a_2 t^3 + (2a_1 + a_2) t^2 + (a_1 + 2a_0) t + a_0) dt = 0$$

$$\frac{2a_2}{4} + \frac{2a_1 + a_2}{3} + \frac{a_1 + 2a_0}{2} + a_0 = 0$$

$$6a_2 + 8a_1 + 4a_2 + 6a_1 + 12a_0 + 12a_0 = 0$$

$$14a_1 + 10a_2 + 24a_0 = 0$$

$$5a_2 + 7a_1 + 12a_0 = 0$$

$$\text{DOF} = 3 - 1 = 2 \Rightarrow$$

$$a_1 = \alpha, a_2 = \beta$$

$$a_0 = -\frac{5\beta}{12} - \frac{7}{12}\alpha$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \alpha \begin{bmatrix} -7/12 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -5/12 \\ 0 \\ 1 \end{bmatrix}$$

$$= \alpha_1 \begin{bmatrix} -7 \\ 12 \\ 0 \end{bmatrix} + \beta_1 \begin{bmatrix} -5 \\ 0 \\ 12 \end{bmatrix}$$

COMPLETE NOTES OF 4/10/23

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} i \\ 1 \end{bmatrix} = 0$$

11/10/23

Adjoint of an operator

$$\mathcal{H} = \frac{d^2}{dx^2} + x \frac{d}{dx}$$

$$x=0 \quad \frac{du}{dx} = 0$$

$$x=1 \quad \frac{du}{dx} + u = 0$$

$$\underline{\underline{A}} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \underline{\underline{u}} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\downarrow \text{operator}$$

$$\underline{\underline{A}} \underline{\underline{u}} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} a_{11}u_1 + a_{12}u_2 \\ a_{21}u_1 + a_{22}u_2 \end{bmatrix}; \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{aligned} \downarrow \text{operator on } u \\ \langle \underline{\underline{A}} \underline{\underline{u}}, \underline{\underline{v}} \rangle &= a_{11}u_1v_1 + a_{12}u_2v_1 + a_{21}u_1v_2 + a_{22}u_2v_2 \\ &= u_1 [a_{11}v_1 + a_{21}v_2] + u_2 [a_{12}v_1 + a_{22}v_2] \end{aligned} \quad \downarrow \text{Rearranging}$$

Then inner product with another vector v , similar to u

$$\langle \underline{\underline{A}} \underline{\underline{u}}, \underline{\underline{v}} \rangle = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} a_{11}v_1 + a_{21}v_2 \\ a_{12}v_1 + a_{22}v_2 \end{bmatrix} \right\rangle$$

$$= \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle$$

$$\langle \underline{\underline{A}} \underline{\underline{u}}, \underline{\underline{v}} \rangle = \langle \underline{\underline{u}}, \underline{\underline{B}} \underline{\underline{v}} \rangle \quad \hookrightarrow \text{Transpose of } A$$

adjoint of matrix operator (Not adjoint of matrix)

For 2×2 matrix or $n \times n$ matrix $\rightarrow$ adjoint operator is transpose

square matrix

But what if matrix is non square?

$$\underline{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

→ If you are operating then operation should be compatible, so u has to be 3×1

$$\underline{A}\underline{u} = \begin{bmatrix} a_{11}u_1 + a_{12}u_2 + a_{13}u_3 \\ a_{21}u_1 + a_{22}u_2 + a_{23}u_3 \end{bmatrix}$$

→ u, v should belong to same space
so 3×1 (v) won't be able to multiply

imp

To get adjoint operator → We have to take complex conjugate of transpose

↓
Here we solved for real values & assume conjugate remains same, so not taken
If would have been complex then would have to take

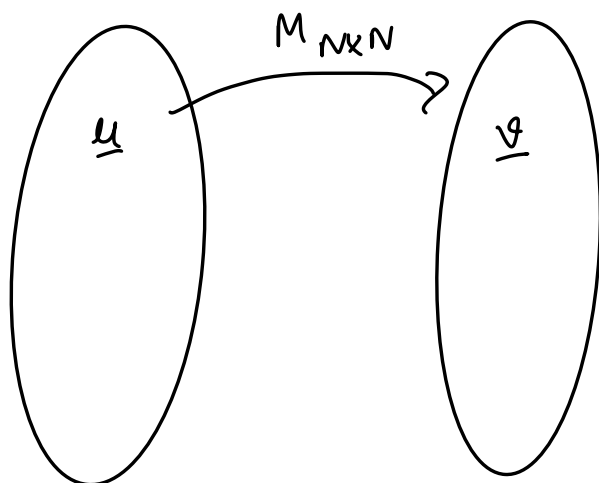
$$\langle \underline{A}\underline{u}, \underline{v} \rangle = \langle \underline{u}, \underline{A}\underline{v} \rangle \rightarrow \text{then } \underline{A} \text{ is self adjoint}$$

If both same then self adjoint

[operate on u , take inner product with v
" " v , " " " u

1. The eigenvalues of self adjoint operator are always real

$$\langle \underline{A}\underline{u}, \underline{u} \rangle = \langle \lambda \underline{u}, \underline{u} \rangle$$



solving factor

$$\underline{A}\underline{u} = \lambda \underline{u} \rightarrow \text{eigenvector}$$

↓
eigenvalue

λ is eigenvalue of A for eigenvector u

after multiplication direction is conserved, only scaled

2. The eigenvectors of a self-adjoint operator corresponding to distinct eigenvalues are always orthogonal

$$\begin{array}{l} \lambda_1 \rightarrow \underline{v_1} \\ \lambda_2 \rightarrow \underline{v_2} \end{array} \quad \left| \quad \begin{array}{l} \langle \underline{A v_1}, \underline{v_2} \rangle = \lambda_1 \langle \underline{v_1}, \underline{v_2} \rangle \\ \langle \underline{v_1}, \underline{A v_2} \rangle = \lambda_2 \langle \underline{v_1}, \underline{v_2} \rangle \end{array} \right.$$

$$0 = (\lambda_1 - \lambda_2) \langle \underline{v_1}, \underline{v_2} \rangle$$

$\hookrightarrow \lambda_1, \lambda_2$ are different so
 $\langle \underline{v_1}, \underline{v_2} \rangle = 0 \rightarrow$ eigenvectors
 are orthogonal
 of self adjoint

Fredholm's alternative theorem

Q Determine the range space for following set of equations

$$\begin{array}{l} x_1 + x_2 + x_3 = 1 \\ 2x_1 + 3x_2 + x_3 = 2 \\ 5x_1 + 6x_2 + 4x_3 = 1 \end{array}$$

For range space $\rightarrow$ you won't
 care about
 constants instead you
 focus on b_1, b_2, b_3

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 2 & 3 & 1 & b_2 \\ 5 & 6 & 4 & b_3 \end{array} \right]$$

$\downarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & -1 & b_2 - 2b_1 \\ 0 & 1 & -1 & b_3 - 5b_1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & -1 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_2 - 3b_1 \end{array} \right]$$

$$b_3 - b_2 - 3b_1 = 0$$

$$b_1 = \alpha \quad b_2 = \beta \quad b_3 = 3\alpha + \beta$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

these will always be linearly independent

$\underline{Ax} = \underline{b}$ is solvable

if $\langle \underline{b}, \underline{y} \rangle = 0$

where $\underline{A}^* \underline{y} = 0$

($\underline{y}$ becomes null space for adjoint of A)

$\Rightarrow \underline{y}$ null space of A^*

$\rightarrow \underline{Ax} = \underline{b}$ is solvable, if $\langle \underline{b}, \underline{y} \rangle = 0$
where $\underline{A}^* \underline{y} = 0$

Proof: $\underline{y}$ is null space of A^* and if $\langle \underline{b}, \underline{y} \rangle = 0$ then we can get range space

Q Determine the null space of $A^* \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 6 \\ 1 & 1 & 4 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 + 2x_2 + 5x_3 = 0$$

$$x_2 + x_3 = 0$$

$$\text{DOF} = 1 \quad [V-E]$$

$$x_2 = \alpha \quad x_3 = -\alpha \quad x_1 = 3\alpha$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} = \underline{y}$$

$$\underline{y} = \alpha \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} \quad \underline{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Null space $\rightarrow$ will solve the homogeneous

Range space $\rightarrow$ that space for which we will have a solution

$$\langle b, y \rangle = 3b_1 + b_2 - b_3 = 0$$

$$b_1 = \beta \quad b_2 = \gamma \quad b_3 = 3\beta + \gamma$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \beta \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

18/10/23

Eigenvalue Problem

Solve the simultaneous equations

$$(a) \quad \frac{dx}{dt} = 2x \rightarrow \text{not coupled}$$

$$\frac{dy}{dt} = -3y$$

$$(b) \quad \frac{dx}{dt} = x + 2y \rightarrow \text{coupled}$$

$$\frac{dy}{dt} = y$$

$$(a) \quad x(t) = c_1 e^{2t}$$

$$y(t) = c_2 e^{-3t}$$

these become easy as although they are simultaneous equations, they are independent

$$(b) \quad y = c_1 e^t$$

$$\frac{dx}{dt} = x + 2c_1 e^t$$

$$\frac{dx}{dt} - x = 2c_1 e^t$$

$$e^{-t} \frac{dx}{dt} - x e^{-t} = 2c_1$$

$$d(xe^{-t}) = 2c_1 dt$$

$$xe^{-t} = 2c_1 t + c_2$$

$$x(t) = 2c_1 t e^t + c_2 e^t$$

→ In this question, it was possible to sequentially solve the coupled equation

$$(a) \quad \frac{dx}{dt} = 2y$$

$$\frac{dy}{dt} = -3x$$

$$(b) \quad \frac{dx}{dt} = x + 2y$$

$$\frac{dy}{dt} = x$$

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\frac{d\underline{x}}{dt} = \underline{A} \underline{x}$$

$$\lambda_1 = -1$$

$$\underline{v}_1 = \begin{bmatrix} 1 & -1 \end{bmatrix}^T$$

$$\lambda_2 = 2$$

$$\underline{v}_2 = \begin{bmatrix} 2 & 1 \end{bmatrix}^T$$

If you have to differentiate a matrix you differentiate every element of it, same for integration

→ Determine the eigenvalue and the "corresponding" eigenvector

→ only take corresponding eigenvector

$$\underline{A} \underline{v}_1 = \lambda_1 \underline{v}_1 \quad \rightarrow \text{eigenvalue}$$

$$\underline{x}(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

↓
corresponding eigenvector

$$x = c_1 e^{-t} + 2c_2 e^{2t}$$

$$y = -c_1 e^{-t} + c_2 e^{2t}$$

$$\underline{x}(t) = \sum_{i=1}^n c_i e^{\lambda_i t} \underline{v}_i$$

eigenvalue × independent variable

↓
corresponding eigenvector

then if our eqns were of form $\frac{d\underline{x}}{dt} = \underline{A} \underline{x}$

$$a) \quad \frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\lambda_1 = 2 \quad \underline{v}_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$$

$$\lambda_2 = -3 \quad \underline{v}_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}^T$$

→ they are "simultaneous" but can be solved independently

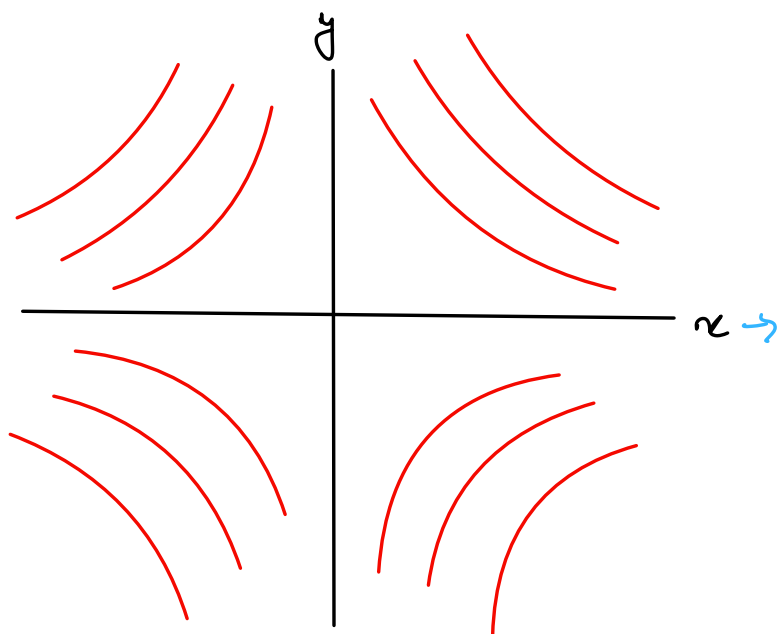


$$\begin{bmatrix} x \\ y \end{bmatrix} = c_1 e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

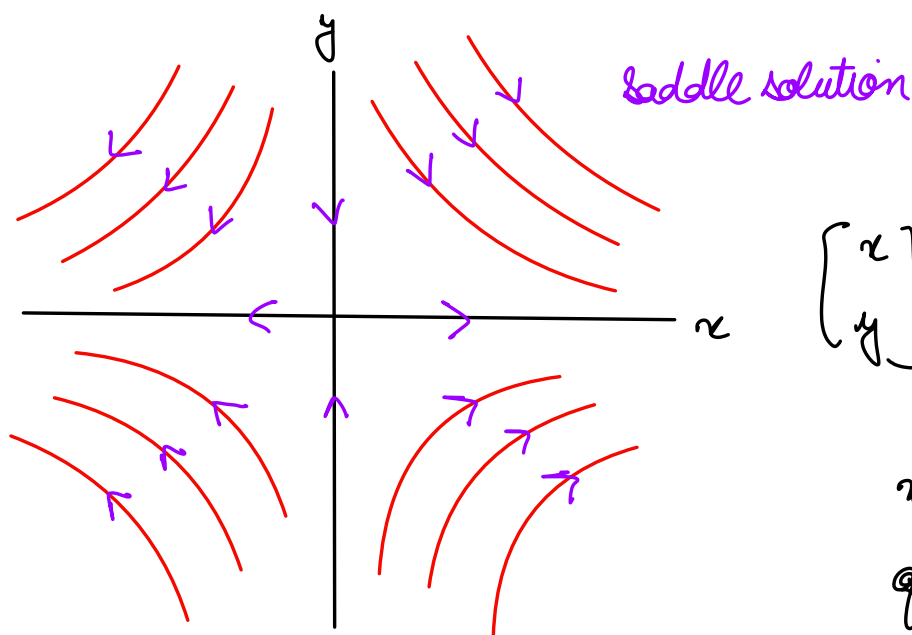
like a parametric equation $\left\{ \begin{array}{l} x = c_1 e^{2t} \rightarrow \text{exponentially } \uparrow \text{ for } \\ y = c_2 e^{-3t} \rightarrow \text{" } \downarrow \text{ for } \end{array} \right.$

both x & y should be satisfied at the same instant of time

$$(c_1 e^{2t}, c_2 e^{-3t})$$



$$\begin{array}{ll} \lambda_1 = 2 & v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T \\ \lambda_2 = -3 & v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \end{array} \left\{ \begin{array}{l} \text{so both } x \text{ \& } y \\ \text{axis are} \\ \text{eigenvectors} \\ \text{for our} \\ \text{problem} \end{array} \right.$$



$$\begin{bmatrix} x \\ y \end{bmatrix} = c_1 e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

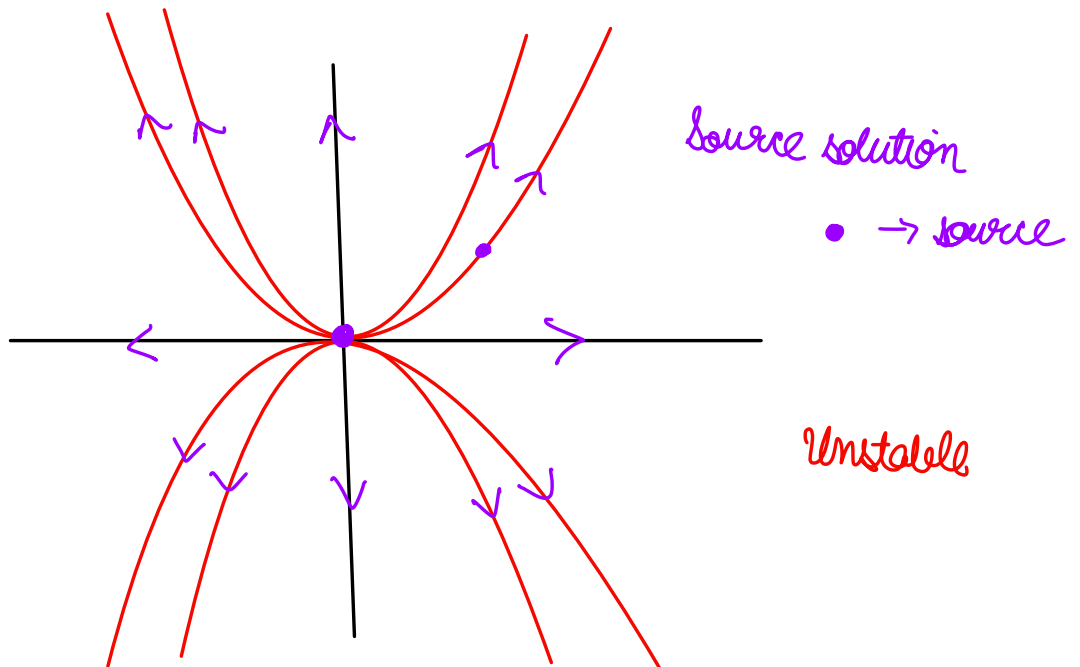
x, y axis are also the solutions of the equation, but are linear combinations of x & y axis solutions

How to make both x & y go simultaneously $0 \rightarrow \infty$

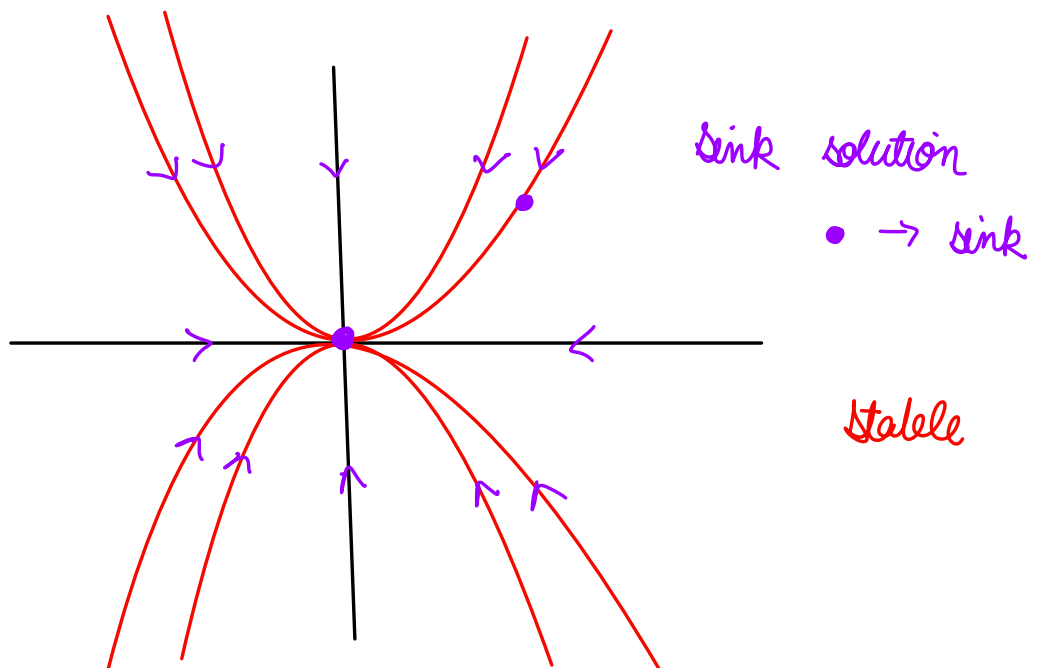
↓

Eigenvalues should change $\rightarrow$ both should become +ve

$$\begin{bmatrix} x \\ y \end{bmatrix} = c_1 e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 e^{+3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



when both eigenvalues are $\rightarrow -ve$



$$a) \quad \frac{d^2 y}{dx^2} - y = 0$$

→ second order equation → but every n^{th} order eqn
can be converted into
 n^{th} order eqn

$$b) \quad \frac{d^2 y}{dx^2} + y = 0$$

$$\text{let } \frac{dy}{dx} = 0y + 1z$$

$$\frac{dz}{dx} = 1y + 0z$$

$$\frac{d}{dx} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\lambda_1 = 1 \quad \underline{v}_1 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$$

$$\lambda_2 = -1 \quad \underline{v}_2 = \begin{bmatrix} 1 & -1 \end{bmatrix}^T$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = c_1 e^x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-x} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$b) \quad \frac{d^2 y}{dx^2} + y = 0$$

$$\frac{d}{dx} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\lambda_1 = i \quad \underline{v}_1 = \begin{bmatrix} -i & 1 \end{bmatrix}^T$$

$$\lambda_2 = -i \quad \underline{v}_2 = \begin{bmatrix} i & 1 \end{bmatrix}^T$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = c_1 e^{ix} \begin{bmatrix} -i \\ 1 \end{bmatrix} + c_2 e^{-ix} \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$= c_1 (\cos x + i \sin x) \begin{bmatrix} -i \\ 1 \end{bmatrix} + c_2 (\cos x - i \sin x) \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -ic_1 \cos x + c_1 \sin x + c_2 i \cos x + c_2 \sin x \\ c_1 \cos x + i c_1 \sin x + c_2 \cos x + c_2 i \sin x \end{bmatrix}$$

$$= \begin{bmatrix} \sin x (c_1 + c_2) + i \cos x (c_2 - c_1) \\ \cos x (c_1 + c_2) + i \sin x (c_1 + c_2) \end{bmatrix}$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} (c_1 + c_2) \sin x \\ (c_2 + c_1) \cos x \end{bmatrix}}_{Z_1 + i Z_2} + i \underbrace{\begin{bmatrix} (c_2 - c_1) \cos x \\ (c_1 + c_2) \sin x \end{bmatrix}}_{Z_1 + i Z_2}$$

$$\frac{d^2}{dx^2} (Z_1 + i Z_2) + (Z_1 + i Z_2) = 0$$

$$\frac{d^2 Z_1}{dx^2} + Z_1 + i \left(\frac{d^2 Z_2}{dx^2} + Z_2 \right) = 0$$

The real part is the solution
The imaginary part is also the solution

↳ so z_1, z_2 both solution

$$\begin{bmatrix} y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} (c_1 + c_2) \sin x \\ (c_2 + c_1) \cos x \end{bmatrix}}_{\text{soln}} + i \underbrace{\begin{bmatrix} (c_2 - c_1) \cos x \\ (c_1 + c_2) \sin x \end{bmatrix}}_{\text{soln}}$$

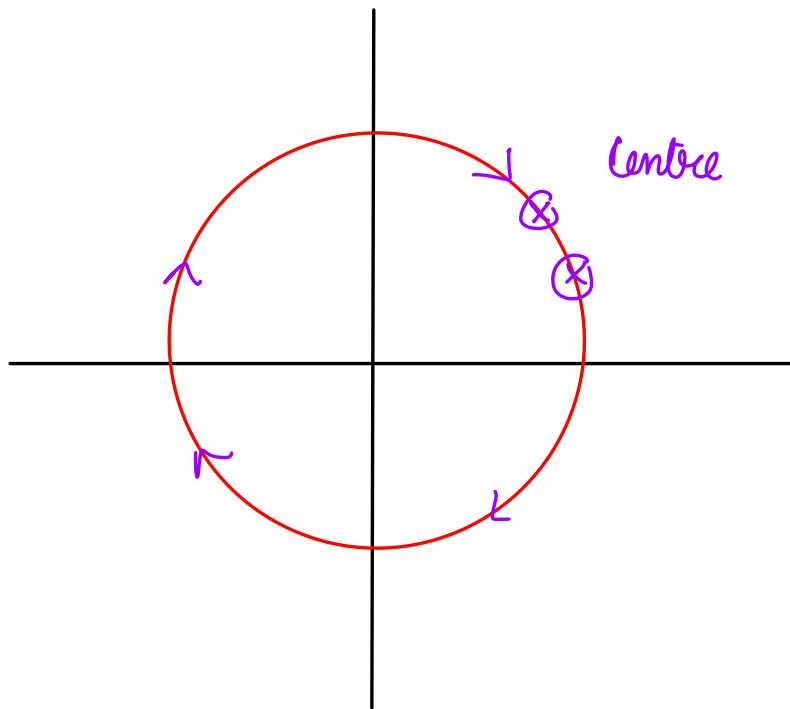
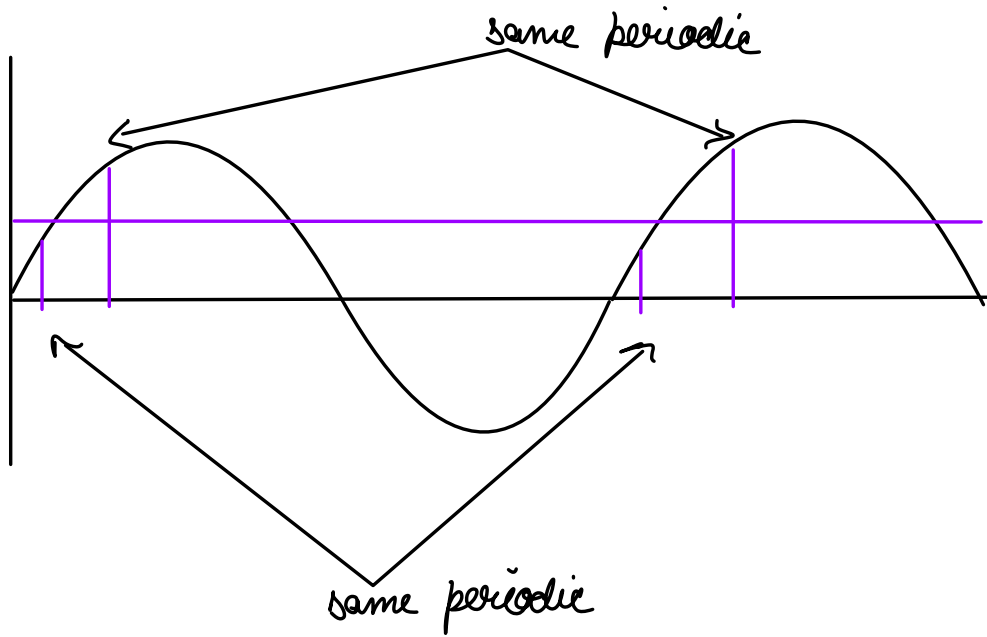
$$\begin{bmatrix} y \\ z \end{bmatrix} = A \begin{bmatrix} \sin x \\ \cos x \end{bmatrix} + B \begin{bmatrix} \cos x \\ -\sin x \end{bmatrix}$$

→ linear combination of real & imaginary part

$$y = \alpha \sin x + \beta \cos x$$

$$z = \alpha \cos x - \beta \sin x$$

} again parametric



Centre solutions are periodic

↓
correspond to system with oscillatory behaviour

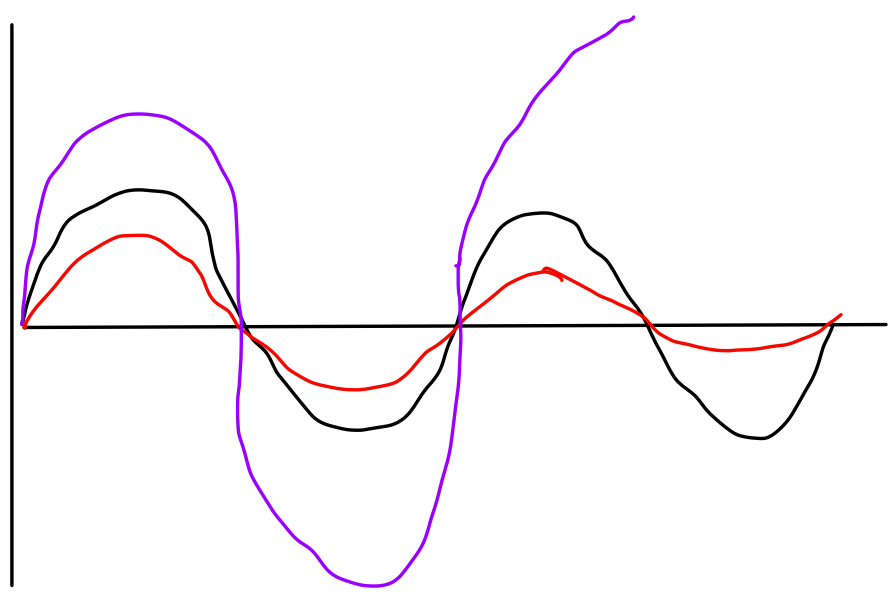
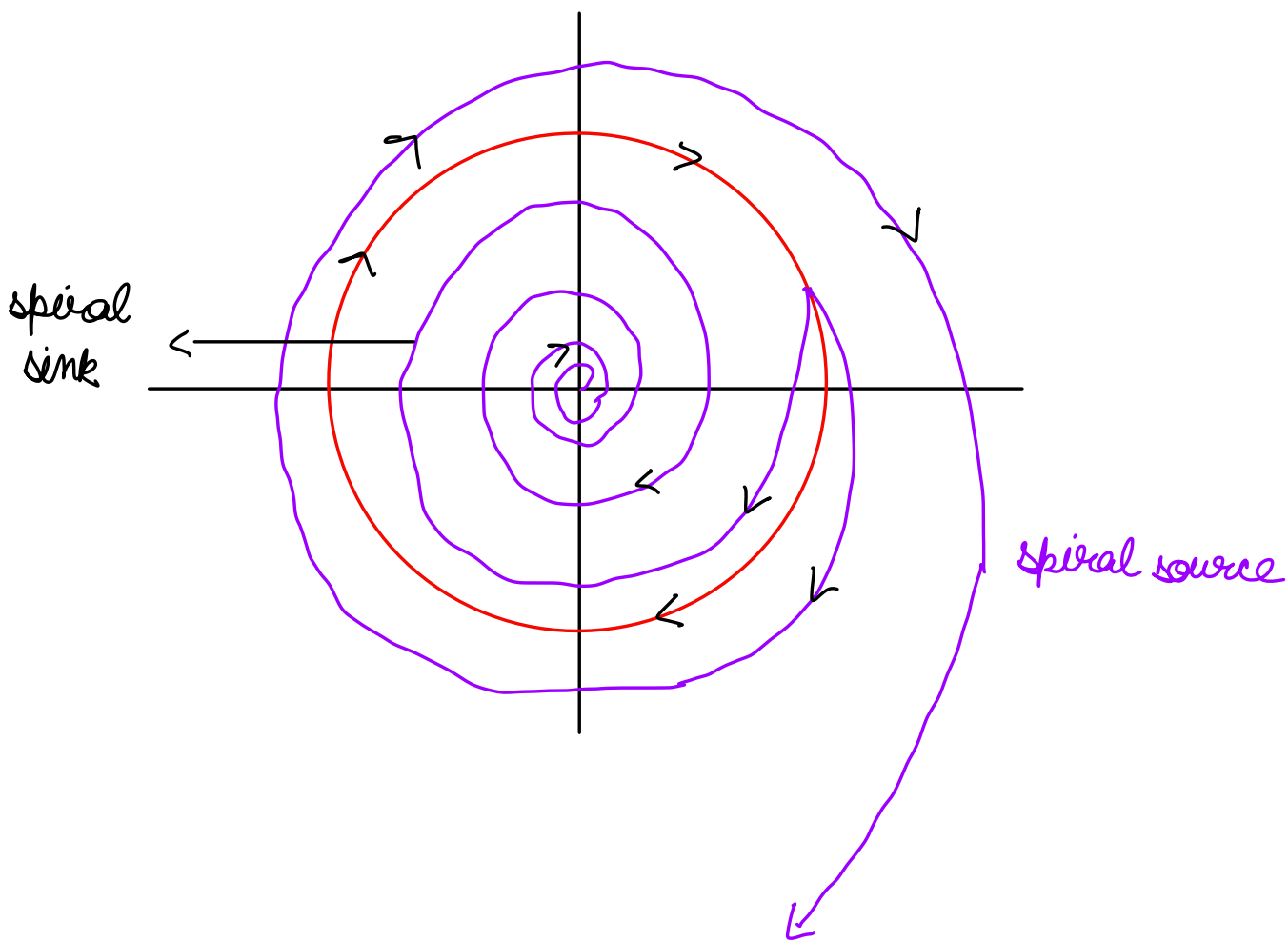
↓
which happens when we have imaginary eigenvalues

In real eigenvalues $\rightarrow$ monotonic behaviour

$$2+i, 2-i, -2-i, -2+i$$

$$e^{(2+i)t}$$

$$e^{2t} \cdot e^{it}$$



$$(1) \quad \frac{dx}{dt} = x + y - t - 1$$

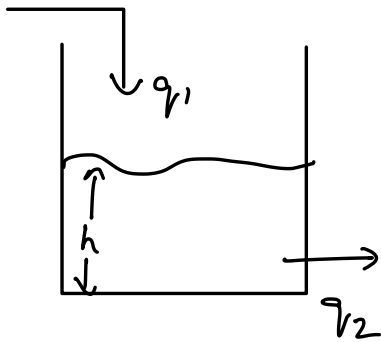
$$\frac{dy}{dt} = 4x + y - 4t - 2$$

$$(2) \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0 \quad n \geq 0$$

$$\frac{dx}{dt} = \underline{A} x$$

$$x(t) = \sum_{i=1}^n c_i e^{\lambda_i t} \underline{v}_i$$

Autonomous $\rightarrow$ govern themselves,
you can't control or vary
things

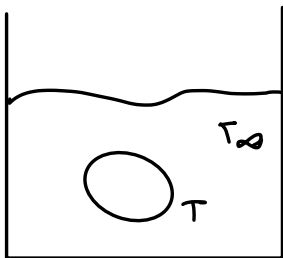


$$\frac{dh}{dt} = \frac{1}{A} (q_1 - q_2)$$

$$\frac{dh}{dt} = -\frac{1}{A} q_2$$

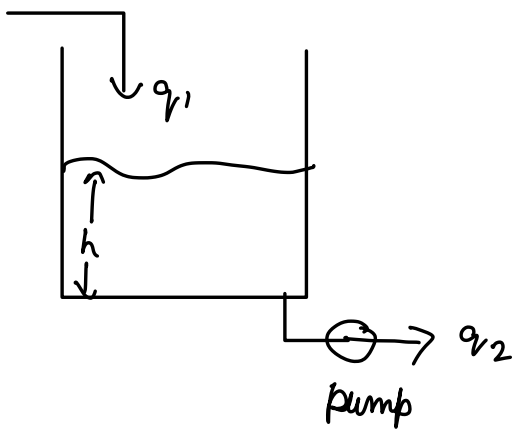
$$\frac{dh}{dt} = \frac{1}{A} f(h)$$

$$\frac{dh}{dt} = \frac{1}{A} d \sqrt{2gh}$$



$$\frac{dT}{dt} = \alpha (T - T_\infty)$$

$\hookrightarrow T_\infty(t) \rightarrow$ can be this to
make it non autonomous



$$\frac{dh}{dt} = -\frac{1}{A} q_2 - b_2(t)$$

$\hookrightarrow$ fn. of t also, after
introduction of pump,
can be controlled

Similarity Transformation

$$\frac{dx}{dt} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} x + \begin{bmatrix} -t-1 \\ -4t-2 \end{bmatrix}$$

$$\frac{dx}{dt} = \underline{A}x + \underline{B}(t)$$

$$\underline{A} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$

$$\lambda_1 = -1 \quad v_1 = [1 \ -2]^T$$

$$\lambda_2 = 3 \quad v_2 = [1 \ 2]^T$$

$$\underline{P} = \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix} \quad \text{augmented matrix of eigenvectors } v_1, v_2$$

$$\underline{P}^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}$$

$$\underline{P}^{-1} \underline{A} = \frac{1}{4} \begin{bmatrix} -2 & 1 \\ 6 & 3 \end{bmatrix}$$

$$\underline{P}^{-1} \underline{A} \underline{P} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} \rightarrow \text{Diagonal matrix with eigenvalues lying on diagonal} \\ [v_1 \ v_2] \text{ if } [v_2 \ v_1] \text{ would have gotten} \\ \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$$

Similar matrices $\rightarrow$ same eigenvalues, but different eigenvectors

$$\frac{dx}{dt} = ax + b(t)$$

$$\frac{dx}{dt} - ax = b(t)$$

$$e^{-at} \frac{dx}{dt} - axe^{-at} = e^{-at} b(t)$$

$$\frac{d(e^{-at}x)}{dt} = e^{-at}b(t)$$

$$\int d(e^{-at}x) = \int e^{-at}b(t) dt$$

$$e^{-at}x = \int e^{-at}b(t) dt + C$$

$$x(t) = e^{at} \int e^{-at}b(t) dt + C e^{at}$$

$$\frac{dx}{dt} = \underline{A}x + \underline{B}(t)$$

$$\frac{dx}{dt} - \underline{A}x = \underline{B}(t)$$

$$\frac{d(\underline{P}^{-1}x)}{dt} - \underline{P}^{-1}\underline{A}x = \underline{P}^{-1}\underline{B}(t)$$

$$\frac{d(\underline{P}^{-1}x)}{dt} - (\underline{P}^{-1}\underline{A}\underline{P})(\underline{P}^{-1}x) = \underline{P}^{-1}\underline{B}(t)$$

$$\frac{dy}{dt} - \underline{\Lambda}y = \underline{G}(t)$$

$$y = \underline{P}^{-1}x$$

$$\underline{\Lambda} = \underline{P}^{-1}\underline{A}\underline{P}$$

$$\underline{G}(t) = \underline{P}^{-1}\underline{B}(t)$$

$$\frac{dx}{dt} - \underline{A}x = \underline{B}(t)$$

$$\frac{dx}{dt} - ax = b(t)$$

$$\left(\frac{dy}{dt}\right)e^{-\underline{\Lambda}t} - \underline{\Lambda}e^{-\underline{\Lambda}t}y = e^{-\underline{\Lambda}t}\underline{G}(t)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^{-\underline{\Lambda}t} = \underline{I} + (-\underline{\Lambda}t) + \frac{1}{2!}(-\underline{\Lambda}t)^2 + \dots$$

$$-\underline{A}t = \begin{bmatrix} t & 0 \\ 0 & -3t \end{bmatrix}$$

$$(-\underline{A}t)^2 = \begin{bmatrix} +t^2 & 0 \\ 0 & (-3t)^2 \end{bmatrix}$$

$$(-\underline{A}t)^3 = \begin{bmatrix} t^3 & 0 \\ 0 & (-3t)^3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -t & 0 \\ 0 & 3t \end{bmatrix} + \begin{bmatrix} (-t)^2 & 0 \\ 0 & (3t)^2 \end{bmatrix} + \begin{bmatrix} (-t)^3 & 0 \\ 0 & (3t)^3 \end{bmatrix} + \dots$$

$$= \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{3t} \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} e^{-\underline{A}t} \underline{y} \\ \underline{y} \end{bmatrix} = e^{-\underline{A}t} G(t)$$

$$d(e^{-\underline{A}t} \underline{y}) = e^{-\underline{A}t} G(t) dt$$

$$\underline{y} = e^{\underline{A}t} \int e^{-\underline{A}t} G(t) dt + e^{\underline{A}t} \underline{c}$$

↳ $\underline{c}$ is a vector as for x there will be a initial condition, for y there will " " "

Sturm-Liouville theory - Bessel Equation : Frobenius Method

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0 \quad n \geq 0$$

$$y = \sum_{i=0}^{\infty} a_i x^{i+k}$$

$$\frac{dy}{dx} = \sum_{i=0}^{\infty} a_i (i+k) x^{i+k-1}$$

$$\frac{d^2 y}{dx^2} = \sum_{i=0}^{\infty} a_i (i+k) (i+k-1) x^{i+k-2}$$

$$\sum_{i=0}^{\infty} a_i (i+k) (i+k-1) x^{i+k} + \sum_{i=0}^{\infty} a_i (i+k) x^{i+k} - n^2 \sum_{i=0}^{\infty} a_i x^{i+k} + \sum_{i=0}^{\infty} a_i x^{i+k+2} = 0$$

$$\Rightarrow \sum_{i=0}^{\infty} a_i [(i+k)(i+k-1) + (i+k) - n^2] x^{i+k} + \sum_{i=0}^{\infty} a_i x^{i+k+2} = 0$$

$x^k \quad x^{k+1} \quad x^{k+2} \quad x^{k+3} \quad \dots$
 this series lagging by two powers this series ahead by two powers

$$a_0 [k(k-1) + k - n^2] = 0$$

$$k = \pm n$$

$$a_1 [k(k+1) + (k+1) - n^2] = 0$$

$$a_1 [k(k+1)^2 - n^2] = 0$$

$$k = 1 \pm n \quad \text{or} \quad a_1 = 0$$

Solving for $k = \pm n$

$$\sum_{i=1}^{\infty} a_i [(i+n)(i+n-1) + (i+n) - n^2] x^{i+n} + \sum_{i=0}^{\infty} a_i x^{i+n+2} = 0$$

$$a_2 \{ (n+2)(n+1) + (n+1) - n^2 \} + a_0 = 0$$

$$a_2 = \frac{-a_0}{2 \cdot 2(n+1)}$$

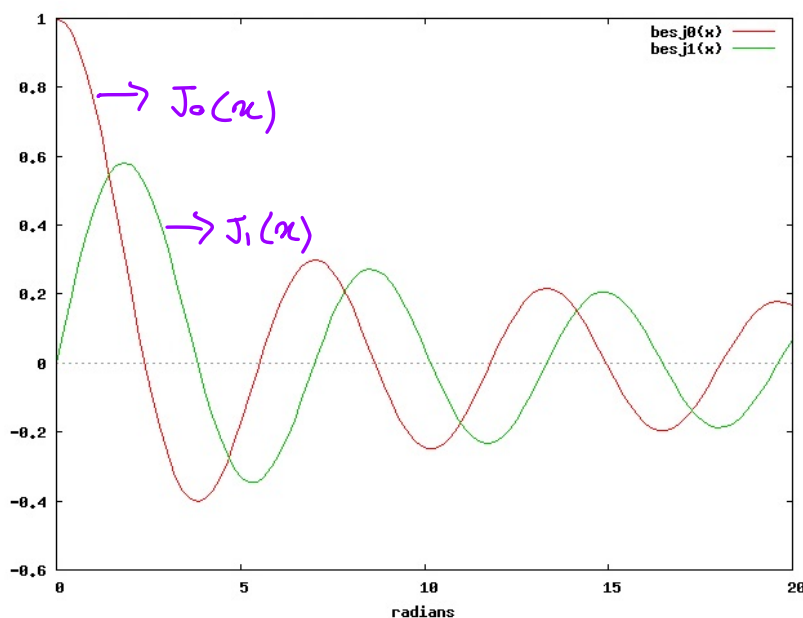
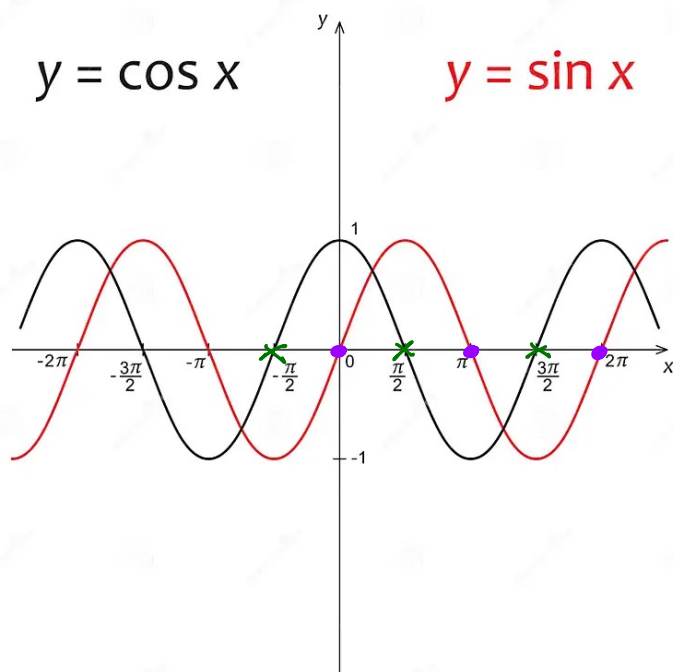
$$a_3, a_4, a_5, a_6$$

$$a_1 = a_3 = a_5 = a_7 \dots = 0 \rightarrow \text{killed half of the series}$$

$$a_4 = \frac{-a_2}{4(2n+4)} = \frac{a_0}{4 \cdot 2 \cdot 2(n+1)(n+2)}$$

$$a_6 = \frac{-a_0}{(n+1)(n+2)(n+3)}$$

We have effectively only 1 unknown a_0



$J_0, J_1, J_2, \dots, J_m \rightarrow$ linearly independent & orthogonal

Bessel fn of second kind $\rightarrow$ obtained by transformation on Bessel fn of first kind

S-L operator

$$\hat{h} = \frac{1}{d\alpha} \left(p(\alpha) \frac{d}{d\alpha} \right) + V(\alpha)$$

- Legendre eqn

- Laguerre eqn

- Hermite eqn